

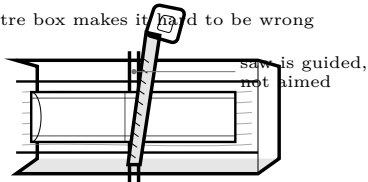
Craft 2: Cut Square

A joint can never be better than the cut that went into it

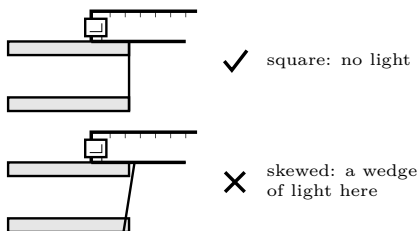
THE STANDARD

The cut face is **perpendicular to the pipe axis** within about one degree, and the whole face is flat. Held against a try square on a flat bench, you should not see light under the blade anywhere. On a 3 in pipe, one degree of skew is about 1.3 mm of gap on the far side — enough to matter, and easy to see once you know to look.

a mitre box makes it hard to be wrong



support the pipe on *both* sides of the cut



Why square matters

A socket joint is a precision taper. A skewed pipe end bottoms on one side while leaving a gap on the other, so the cement pools where it is not wanted and the weld area is smaller than it looks from outside. The joint appears finished and is not.

Use a guide

Freehand sawing a round object square is genuinely difficult and there is no reason to try. A mitre box or a tubing cutter turns a skill problem into an equipment problem.

Tubing cutter

Best result on pipe up to about 2 in — it cuts square by construction. Tighten a quarter turn per revolution, never more, or it ploughs and leaves a lip.

Fine teeth

On plastic, a fine-toothed saw cuts cleanly and a coarse one tears. Let the saw do the work; pushing hard makes the blade wander.

Support both sides

Pipe that drops as the cut finishes tears the last of the material and takes the face out of square in the final centimetre.

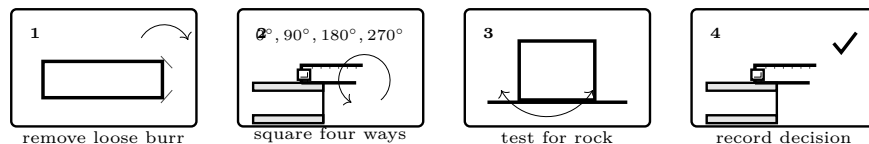
Cut on the waste side

Of the line, so the line survives on the piece you are keeping. *see the **Craft 1: Measure & Mark** sheet*

Method

1. Align the full-circumference mark with the mitre-box slot. **Check:** the line meets the slot on both visible sides. If it does not, return to Craft 1.
2. Clamp or hold the pipe against the fence. **Check:** try to roll it with the saw clear; any movement means the support must be reset.
3. Support the offcut. **Check:** it stays level when pressed lightly. If it drops, add support before the saw touches the pipe.
4. Pull the saw backward twice to establish the kerf. **Check:** the kerf lies on the waste side and the pencil line survives. If the kerf crosses the kept side, stop and remake the part from a new mark.
5. Saw with long strokes and light pressure. **Check:** the blade stays in the guide and the pipe stays against the fence. If either moves, stop the saw completely and reset the work.
6. For pipe that must be turned, remove the stopped saw before rotating the pipe. **Check:** the existing kerf realigns with the guide. If it does not, restore the previous orientation and support rather than starting a second kerf.
7. Finish with light strokes while holding the offcut level. **Check:** the last material severs without tearing. A torn edge must be re-cut from a new full-circumference mark.

Checking, and what to do if it is off



Test	What it tells you
Try square, light behind	The direct test. Rotate the square around the pipe to four positions — a cut can be square in one plane and skewed in the other.
Stand it on glass	A truly square end stands on a flat surface without rocking. Rock it gently; any wobble is skew.
Dry-fit and rotate	Push the pipe into its fitting and rotate it. If the gap at the shoulder changes as it turns, the end is not square.

Inspection decision

First remove only loose burrs; do not reshape the face. Ask: *Is there light at any of four square positions? Does the pipe rock on glass? Does the fitting shoulder gap change as the dry fit turns?*

Pass only if every answer is no and the finished length remains within 1 mm of the cut list. If any answer is yes, **re-cut from a new mark**. Do not sand the face square: it removes an unknown amount and can round the edge. If re-cutting would make the part undersize, **remake the part** from new stock. Keep the rejected piece only as a clearly marked practice coupon.

The habit worth taking away

Every trade has a version of this: use a guide rather than your eye, and then check against a known-good reference rather than against your own judgement of how it looks. A try square is a piece of steel that is more perpendicular than you are. That is not an insult, it is the entire reason it exists.

Final proof record

Finished length	_____ mm	Rocking on glass	no / yes
Light at 0°	no / yes	Light at 90°	no / yes
Light at 180°	no / yes	Light at 270°	no / yes
Rotating shoulder gap	no / yes	Cut-list tolerance	within 1 mm: no / yes

Circle the disposition: **PASS** / **RE-CUT** / **REMAKE**.

Inspector initials _____ Date _____

If not passed, record the failed test and the new cut mark or remake:

Only a recorded **PASS** proceeds to a joint.

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